



Central Banking System

Presented by: Group 4

Md Fuad Anan (251014032)

Tapu Shikari (222014064)

Sithi Saha (242014064)

Jannatul Ferdous Lamia (242014001)

Maha Islam (252014045)

Presented to:

Md Al Amin Sheikh, Ph.D.

Assistant Professor, Department of CSE

University of Liberal Arts Bangladesh

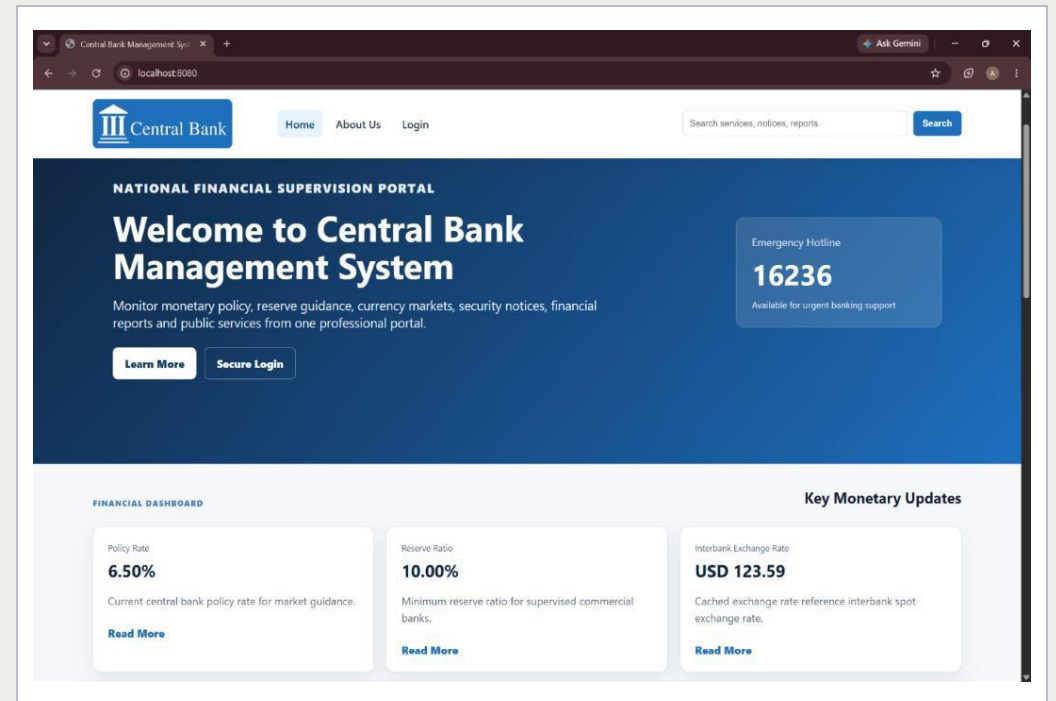
Report date: 19 August 2026



The project solves a monitoring problem

Central-bank supervision becomes difficult when information is scattered:

- Multiple commercial banks must be monitored together
- Loans, reserves, transactions, and approvals need clear records
- Suspicious activity is hard to identify manually
- Reports and decisions need reliable, centralized data





The objective is controlled central-bank supervision

- 01 — Register and manage commercial banks under one central system.
- 02 — Support money supply, liquidity, reserve, and monetary policy operations.
- 03 — Monitor inter-bank transactions and flag suspicious activity.
- 04 — Manage customers, accounts, loans, approvals, and notifications.
- 05 — Generate reports for banks, reserves, fraud, loans, performance, and risk.





Three user roles keep authority separated

Governor

Highest authority
Manages administrator access
Approves or rejects critical requests

Administrator

Submits bank applications
Decides loans
Requests policy, reserve, and supply changes

Commercial Bank

Uses bank dashboard
Applies for loans and transfers
Manages customers, accounts, and profile

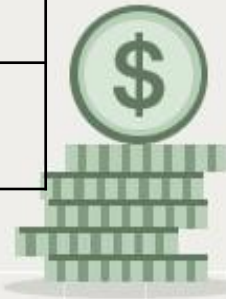
Use-case view from the final report





The feature set covers the main banking workflow

Management	Financial Operations	Monitoring and Output
Bank registration and suspension	Inter-bank fund transfers	Risk assessment
Customer and account handling	Loan applications and decisions	Fraud detection
Role-based dashboards	Deposits and withdrawals	Approval requests
Administrator and Governor control	Reserve management	Notifications
Commercial bank profile control	Monetary policy updates	Audit and activity logs
Data stored in structured CSV files	Exchange-rate information	Bank, loan, reserve, fraud, and summary reports





Architecture and data flow

Browser UI PageRenderer

HTML and CSS pages rendered from Java



AppRouter

Routes GET/POST requests to the correct operation



Services

CentralBankSystem, LoginManager, RiskAssessment, FraudDetector



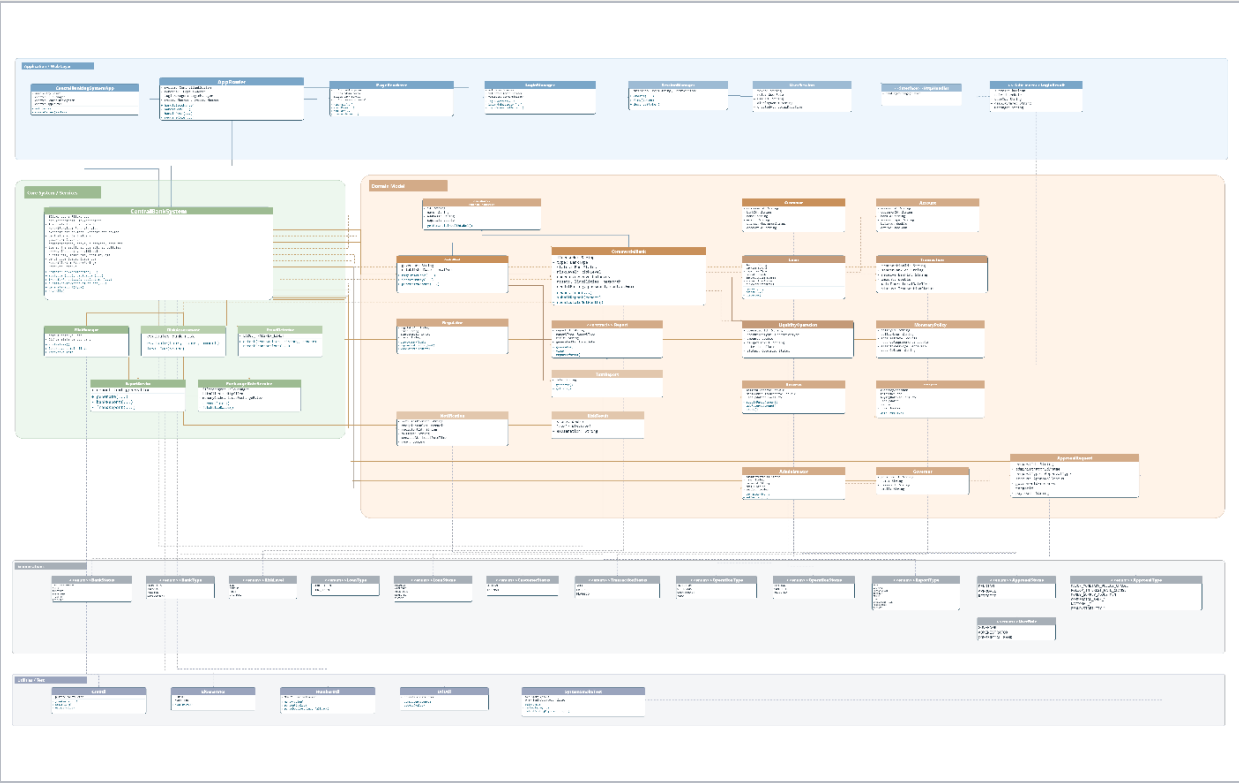
Models

Bank, Customer, Account, Loan, Transaction, Reserve, Policy



CSV Storage

Banks, customers, accounts, loans, transactions, reserves, logs





Risk and approval controls

Risk assessment

- Reserve position
- Debt ratio
- Credit rating
- Previous loan history
- Requested amount vs. capital
- Outstanding loan exposure

Output: score + **LOW / MEDIUM / HIGH / CRITICAL** level

Fraud detection

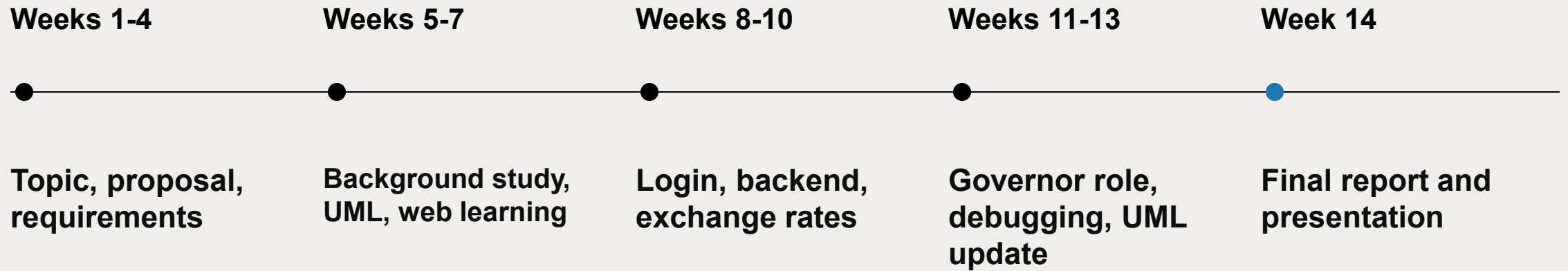
- Zero or negative transfer amount
- Transfer of 500,000 or more
- Negative sender balance
- Repeated failed transfers
- Repeated 100,000+ transfers between same banks (24h)

Output: warnings, **FLAGGED** status, alerts, and audit log





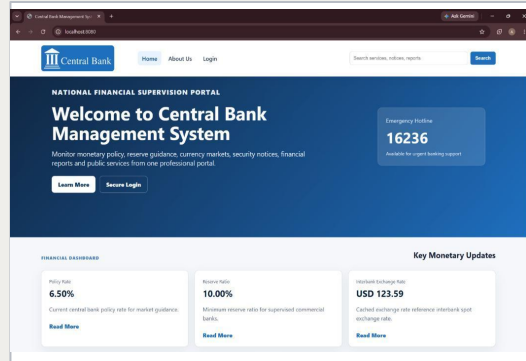
Development followed an iterative path



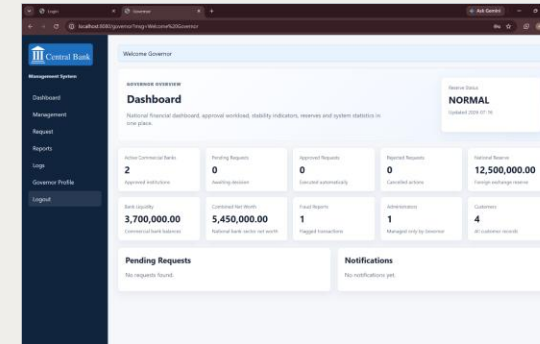


The implemented interface proves the workflow

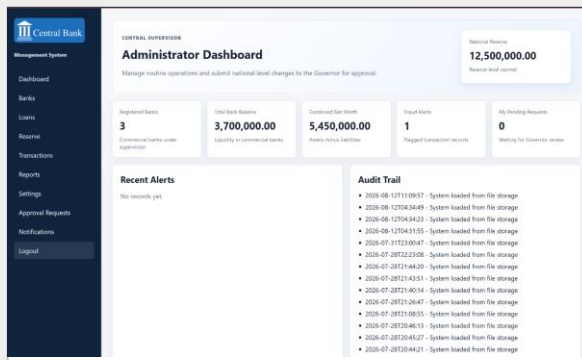
Public home



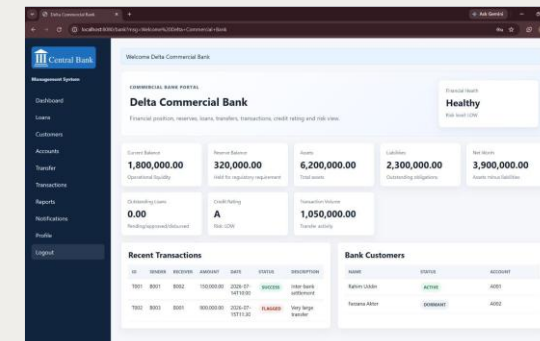
Governor dashboard



Administrator dashboard



Commercial bank dashboard





Conclusion

The simulator is a foundation for a stronger system

Future improvements:

- Relational database integration
- MFA, encryption, and security auditing
- Interactive analytical dashboards
- Machine-learning fraud detection
- Larger-data performance testing

Limitation:

- Simulation only; not a real central banking system
- Simplified financial, risk, and fraud processes
- Uses CSV files instead of a scalable database
- Limited data handling and scalability
- Basic fraud detection and risk analysis
- Weak security and lack of full compliance
- Not suitable for high transaction volume or real-time monitoring





Thank You! & Any questions?

